

Kubernetes Assignment-3

Kubernetes Deployment:

Step 1: Start Minikube

Start the Kubernetes cluster using Docker:

```
minikube start --driver=docker
```

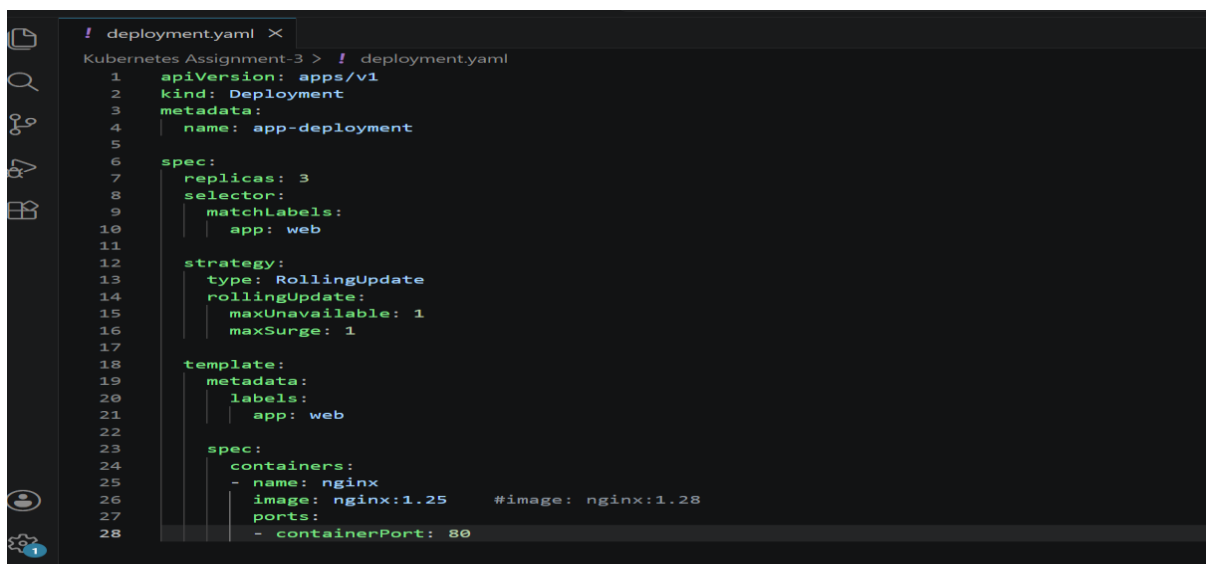
Check the Minikube status:

```
minikube status
```

Step 2: Check the Kubernetes Node

```
kubectl get nodes
```

Step 3: Create the Deployment YAML File



```
! deployment.yaml X
Kubernetes Assignment-3 > ! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    | name: app-deployment
5
6  spec:
7    | replicas: 3
8    | selector:
9    | | matchLabels:
10   | | app: web
11
12   | strategy:
13   | | type: RollingUpdate
14   | | rollingUpdate:
15   | | maxUnavailable: 1
16   | | maxSurge: 1
17
18   | template:
19   | | metadata:
20   | | labels:
21   | | app: web
22
23   | | spec:
24   | | containers:
25   | | - name: nginx
26   | | image: nginx:1.25 #image: nginx:1.28
27   | | ports:
28   | | - containerPort: 80
```

Step 4: Create the Deployment

Apply the YAML file:

```
kubectl apply -f deployment.yaml
```

Step 5: Verify the Deployment

```
kubectl get deployments
```

Step 6: Check the ReplicaSet

kubectl get rs

Step 7: Check the Pods

kubectl get pods

Step 8: Describe the Deployment

kubectl describe deployment app-deployment

This displays detailed information about:

- * Deployment name
- * Namespace
- * Replica count
- * Labels
- * Selector
- * Container image
- * ReplicaSet
- * Events

Step 9: Check the Pod Details

kubectl get pods

Step 9.1: Then describe a Pod:

kubectl describe pod <pod-name>

Step 10: Update the Container Image

Edit `deployment.yaml` and change, Save the file.

Step 11: Apply the Updated Configuration

kubectl apply -f deployment.yaml

Kubernetes detects the change and starts a **rolling update**.

A new ReplicaSet is created for the updated Pod template.

Step 12: Watch the Rolling Update

kubectl get pods -w

The `-w` option means **watch**.

It continuously displays changes as old Pods are replaced by new Pods.

Step 13: Check Rollout Status

kubectl rollout status deployment app-deployment

Step 14: Check Rollout History

kubectl rollout history deployment app-deployment

Each revision represents a different version of the Deployment configuration.

Step 15: View a Specific Revision

kubectl rollout history deployment app-deployment --revision=1

This allows you to inspect the configuration stored for that revision.

Step 16: Roll Back the Deployment

If the new version has a problem, roll back to the previous revision:

kubectl rollout undo deployment app-deployment

Or roll back to a specific revision:

kubectl rollout undo deployment app-deployment --to-revision=1

Step 17: Verify the Rollback

Check the rollout:

kubectl rollout status deployment app-deployment

The Pod should now use the image from the rolled-back revision.

Deployment responsibilities

- * Creates and manages ReplicaSets.
- * Maintains the desired number of Pods.
- * Performs rolling updates.
- * Supports rollbacks.
- * Maintains application availability.

```
File Edit Selection View Go Run ... Kubernetes Assignments Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl apply -f deployment.yaml
deployment.apps/app-deployment unchanged

Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
app-deployment-9f465bb9c-65m7c      1/1     Running   0           6m9s
app-deployment-9f465bb9c-9mf89      1/1     Running   0           6m8s
app-deployment-9f465bb9c-f952w      1/1     Running   0           5m16s
mynginx                             1/1     Running   2 (29m ago) 4d
replicademo-rs-47zdt                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-98pmj                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-k92rb                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-nzvvg                1/1     Running   2 (29m ago) 3d21h

Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl describe deployment app-deployment
Name:                               app-deployment
Namespace:                           default
CreationTimestamp:                   Fri, 07 Aug 2026 17:05:58 +0530
Labels:                               <none>
Annotations:                         deployment.kubernetes.io/revision: 2
Selector:                             app=web
Replicas:                             3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:                         RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:      nginx:1.28
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl describe deployment app-deployment
Name:                               app-deployment
Namespace:                           default
CreationTimestamp:                   Fri, 07 Aug 2026 17:05:58 +0530
Labels:                               <none>
Annotations:                         deployment.kubernetes.io/revision: 1
Selector:                             app=web
Replicas:                             3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:                         RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:      nginx:1.25
      Port:       80/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations: <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  app-deployment-c77955c54 (3/3 replicas created)
```

```
File Edit Selection View Go Run ... Kubernetes Assignments Update
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl apply -f deployment.yaml
deployment.apps/app-deployment unchanged

Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
app-deployment-9f465bb9c-65m7c      1/1     Running   0           6m9s
app-deployment-9f465bb9c-9mf89      1/1     Running   0           6m8s
app-deployment-9f465bb9c-f952w      1/1     Running   0           5m16s
mynginx                             1/1     Running   2 (29m ago) 4d
replicademo-rs-47zdt                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-98pmj                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-k92rb                1/1     Running   2 (29m ago) 3d21h
replicademo-rs-nzvvg                1/1     Running   2 (29m ago) 3d21h

Acere@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl describe deployment app-deployment
Name:                               app-deployment
Namespace:                           default
CreationTimestamp:                   Fri, 07 Aug 2026 17:05:58 +0530
Labels:                               <none>
Annotations:                         deployment.kubernetes.io/revision: 2
Selector:                             app=web
Replicas:                             3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:                         RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:      nginx:1.28
```

```
Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl rollout history deployment app-deployment
deployment.apps/app-deployment
REVISION  CHANGE-CAUSE
2          <none>
3          <none>

Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl rollout undo deployment app-deployment
Warning: resource deployments/app-deployment was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/app-deployment rolled back

Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl get pods -w
NAME                                READY  STATUS   RESTARTS   AGE
app-deployment-9f465bb9c-9tg4w     1/1    Running  0          94s
app-deployment-9f465bb9c-mpck4     1/1    Running  0          99s
app-deployment-9f465bb9c-pd9jt     1/1    Running  0          99s
mynginx                             1/1    Running  2 (47m ago) 4d
replicademo-rs-47zdt               1/1    Running  2 (47m ago) 3d22h
replicademo-rs-98pmj               1/1    Running  2 (47m ago) 3d22h
replicademo-rs-k92rb               1/1    Running  2 (47m ago) 3d22h
replicademo-rs-nzvvg               1/1    Running  2 (47m ago) 3d22h
```

```
Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl rollout status deployment app-deployment
deployment "app-deployment" successfully rolled out

Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl describe deployment app-deployment
Name:                app-deployment
Namespace:            default
CreationTimestamp:    Fri, 07 Aug 2026 17:05:58 +0530
Labels:               <none>
Annotations:          deployment.kubernetes.io/revision: 4
Selector:             app=web
Replicas:             3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:         RollingUpdate
MinReadySeconds:      0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:        nginx:1.28
      Port:         80/TCP
      Host Port:    0/TCP
      Environment:  <none>
      Mounts:       <none>
      Volumes:      <none>
      Node-Selectors:  <none>
      Tolerations:   <none>
  Conditions:
    Type             Status  Reason
```

```
Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl rollout history deployment app-deployment
deployment.apps/app-deployment
REVISION  CHANGE-CAUSE
3          <none>
4          <none>

Acerg@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
• $ kubectl rollout history deployment app-deployment --revision=3
deployment.apps/app-deployment with revision #3
Pod Template:
  Labels:  app=web
           pod-template-hash=c77955c54
  Containers:
    nginx:
      Image:        nginx:1.25
      Port:         80/TCP
      Host Port:    0/TCP
      Environment:  <none>
      Mounts:       <none>
      Volumes:      <none>
      Node-Selectors:  <none>
      Tolerations:   <none>
```

```

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
● $ kubectl rollout undo deployment app-deployment
Warning: resource deployments/app-deployment was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/app-deployment rolled back

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
● $ kubectl describe deployment app-deployment
Name:          app-deployment
Namespace:     default
CreationTimestamp: Fri, 07 Aug 2026 17:05:58 +0530
Labels:        <none>
Annotations:   deployment.kubernetes.io/revision: 5
Selector:      app=web
Replicas:      3 desired | 3 updated | 3 total | 2 available | 1 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:      nginx:1.25
      Port:       80/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations: <none>
  Conditions:

```

```

$ kubectl describe deployment app-deployment
Normal ScalingReplicaSet 6s (x2 over 22m) deployment-controller Scaled up replica set app-deployment-c77955c54 from 1 to 2
Normal ScalingReplicaSet 1s (x6 over 22m) deployment-controller (combined from similar events): Scaled down replica set app-deployment-9f465bb9c from 1 to 0

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
● $ kubectl rollout history deployment app-deployment
deployment.apps/app-deployment
REVISION  CHANGE-CAUSE
4         <none>
5         <none>

Acer@DESKTOP-KND377Q MINGW64 ~/SRE-DEVOPS/te-5th_and_6th_batch_assignments/Saranya_m/Kubernetes Assignments/Kubernetes Assignment-3 (main)
○ $ kubectl get pods -w
NAME                                READY  STATUS   RESTARTS   AGE
app-deployment-c77955c54-p14rc      1/1    Running  0          43s
app-deployment-c77955c54-plvbz      1/1    Running  0          43s
app-deployment-c77955c54-vf82d      1/1    Running  0          39s
mynginx                             1/1    Running  2 (54m ago)  4d
replicademo-rs-47zdt                1/1    Running  2 (54m ago)  3d22h
replicademo-rs-98pmj                1/1    Running  2 (54m ago)  3d22h
replicademo-rs-k92rb                1/1    Running  2 (54m ago)  3d22h
replicademo-rs-nzvvg                1/1    Running  2 (54m ago)  3d22h

```